



UNIVERSITY OF
CAMBRIDGE

Research Informatics Training

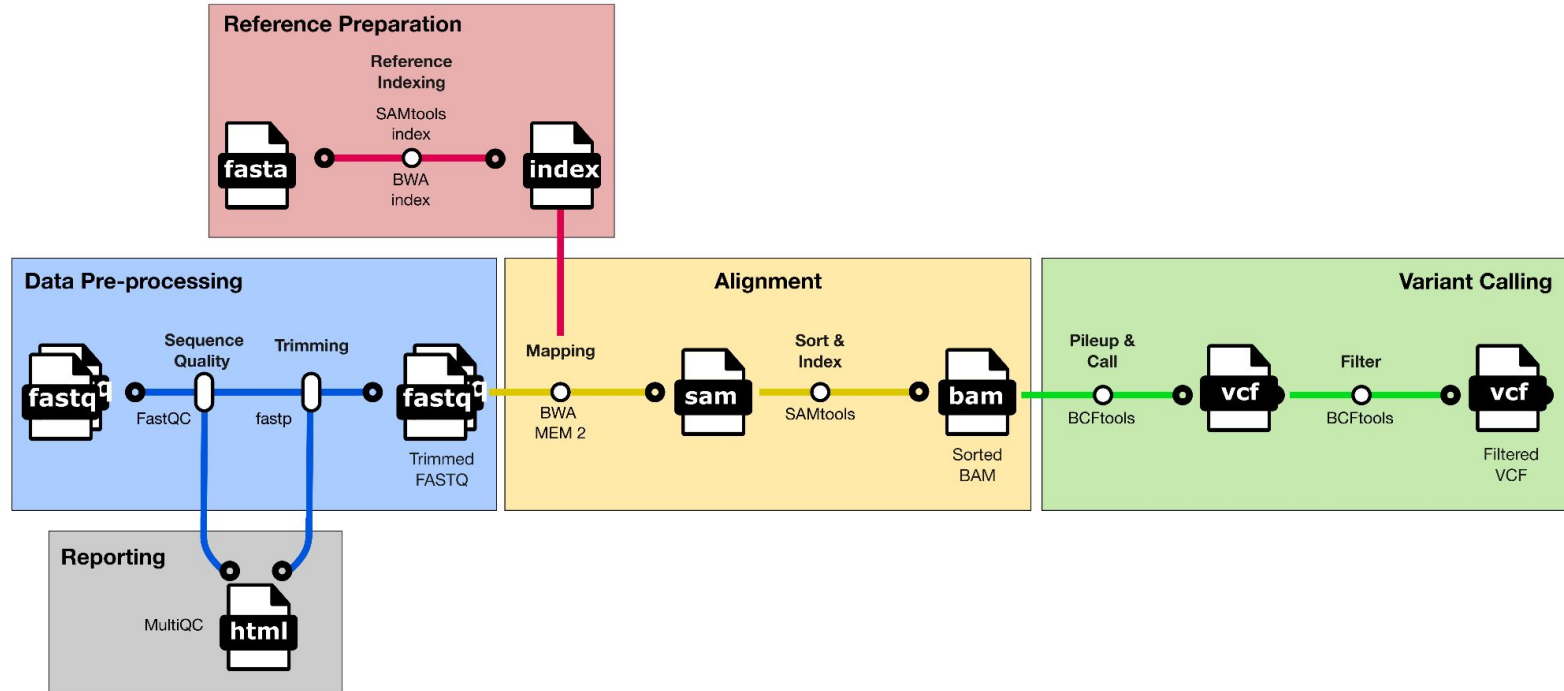
Introduction to NGS Analysis

Standard NGS workflow

The journey of a sample typically follows three main stages:

1. **Library Preparation (Wet Lab):** DNA/RNA is fragmented, and adapters are added to prepare the "library" for the sequencer.
2. **Sequencing (The Machine):** The library is loaded into a flow cell where the actual "base calling" happens, generating raw digital data.
3. **Data Analysis (Dry Lab):** The raw data (FASTQ) is cleaned, aligned to a reference, and analyzed for biological insights (Variants, Gene Expression, etc.).

Overview



Introduction to NGS Analysis

By the end of this course, you will be able to:

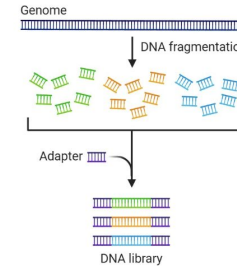
- Navigate the standard **Linux command line** for data processing.
- Understand and manipulate common **genomic file formats** (e.g. FASTQ, BAM, VCF).
- Perform **Quality Control (QC)** to ensure data integrity.
- Transform raw sequencing reads into meaningful **biological results**:
 - Map sequence reads to a reference genome
 - Visualise the mapped reads
 - Call variants

Introduction to NGS technologies

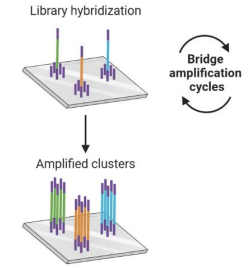
Sequencing: Illumina

- **Library Prep:** DNA is cut into small fragments, and special adapters are attached to the ends.
- **Cluster Generation:** Each fragment is stuck to the flow cell and copied thousands of times in a tiny spot called a **cluster**. This makes the signal bright enough for a camera to see.
- **Sequencing by Synthesis (SBS):** The machine adds fluorescently labeled bases (A, C, G, T) one by one. As each base is added, it emits a unique color of light. A high-resolution camera snaps a photo, and the computer "calls" the base based on the color.

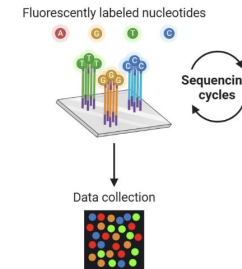
① Library preparation



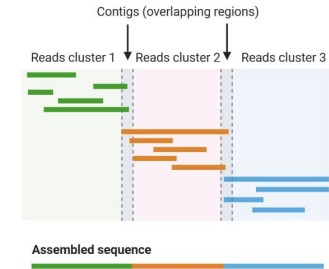
② DNA library bridge amplification



③ DNA library sequencing



④ Alignment and data analysis



Advantages of Illumina sequencing

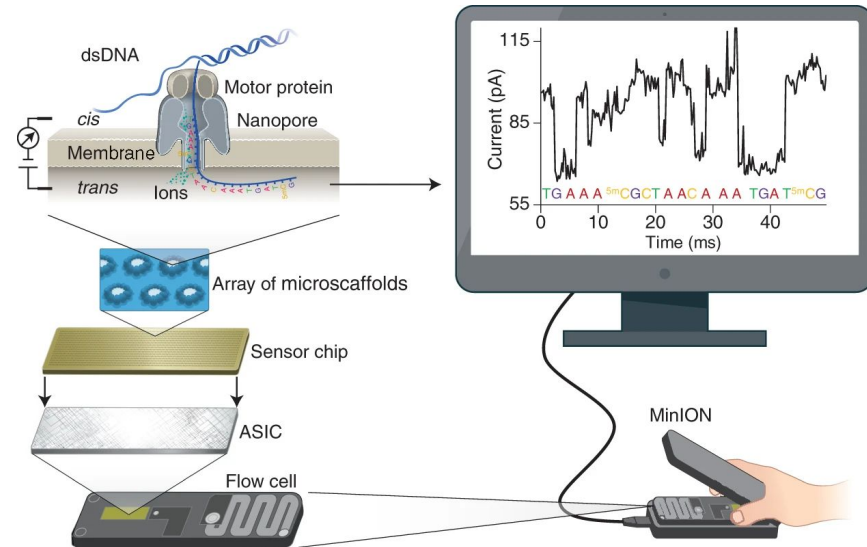
- **Highest Accuracy:** It boasts the lowest "raw" error rate in the industry (typically >Q30, or 99.9% accuracy). This makes it the most trusted platform for clinical diagnostics and finding rare mutations.
- **Massive Throughput:** A single run on a high-end machine (like the NovaSeq) can generate terabytes of data—enough to sequence dozens of human genomes at once.
- **Cost-Effectiveness:** Because it sequences so many fragments at once, the cost per megabase of data is significantly lower than almost any other technology.
- **Mature Ecosystem:** Since it is the industry standard, almost every bioinformatics tool and "kit" (for cancer, ancestry, or COVID tracking) is designed specifically for Illumina data.

Disadvantages of Illumina sequencing

- **The "Jigsaw Puzzle" Problem:** Because the reads are so short, it is difficult to reassemble parts of the genome that are highly repetitive. I
- **Structural Variation Gap:** Large-scale changes in the genome (like big chunks of DNA moving from one chromosome to another) are often missed because the reads aren't long enough to "span" the break points.
- **GC Bias:** The chemical process used (PCR) can struggle with areas of the genome that are very rich in G and C bases, leading to "blind spots" in coverage.
- **Capital Investment:** While the cost *per sample* is low, the machines themselves are incredibly expensive (up to \$1 million), requiring specialized facilities and high-volume use to be economical.

Sequencing: Oxford Nanopore

- **The Nanopore:** A tiny, protein-based pore is embedded in an electrically resistant membrane. A constant current flows through the pore.
- **The Translocation:** An enzyme (a "motor protein") grabs a DNA strand and unzips it, feeding a single strand through the pore.
- **The Signal:** As different bases (A, C, G, T) pass through the pore, they physically block the hole in slightly different ways. This causes a unique "wiggle" or disruption in the electrical current.
- **Basecalling:** Sophisticated AI algorithms (Neural Networks) translate these electrical wiggles (called "squiggles") back into a DNA sequence in real-time.



Advantages of ONT

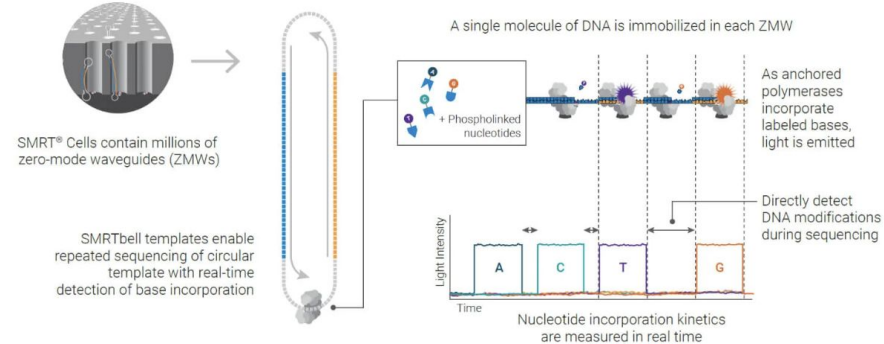
- **Ultra-Long Reads:** While Illumina stops at 300bp, ONT can sequence fragments over **1 million base pairs** long. This makes it the "ultimate puzzle solver" for complex genomes and repetitive regions.
- **Real-Time Analysis:** You can see the data as it is being sequenced. If you find what you're looking for (e.g., a specific pathogen), you can stop the run immediately.
- **Portability (The MinION):** The smallest sequencer is the size of a USB thumb drive. It has been used in the International Space Station, in remote jungles to track Ebola, and in mobile clinics.
- **Direct Modification Detection:** Because it "feels" the DNA, it can detect chemical changes like **methylation** directly, without the need for the harsh chemical treatments required by other platforms.

Disadvantages of ONT

- **Higher Raw Error Rate:** While modern "R10" pores have greatly improved, individual reads still have a higher error rate (~1–5%) than Illumina (<0.1%). This can make it harder to find very rare single-nucleotide mutations.
- **Homopolymer Issues:** The pore struggles when the same base repeats many times (e.g., **AAAAAA**). It is hard for the sensor to tell exactly how many 'A's passed through if the electrical signal doesn't change.
- **High Computational Demand:** Translating raw electrical signals into DNA (basecalling) requires powerful GPUs. You cannot process high-volume Nanopore data efficiently on a standard office laptop.
- **Cost of Scale:** While the "entry-level" MinION is cheap (~\$1,000), the cost *per gigabase* of data on a large scale can sometimes be higher than Illumina's massive factory-scale machines.

Sequencing: Pacific Biosciences

- **The SMRT Cell:** The flow cell contains millions of microscopic wells called **Zero-Mode Waveguides (ZMWs)**. Each ZMW is so small that light cannot fully pass through it.
- **The Movie:** A single DNA polymerase is anchored at the bottom of each ZMW. As it incorporates fluorescently labeled bases, a laser illuminates only the very bottom of the well.
- **The HiFi Loop:** PacBio uses "SMRTbell" adapters to turn linear DNA into a closed circle. The enzyme goes around the circle multiple times. The computer then takes the "consensus" of these multiple passes to cancel out random errors.



Advantages of PacBio

- **Ultra-High Accuracy (99.9%):** By reading the same circular molecule 10–15 times, PacBio produces long reads (15,000 to 20,000 bp) that are just as accurate as Illumina's short reads.
- **Superior Assembly:** Because the reads are both long and accurate, they can easily bridge complex repetitive regions and resolve "haplotypes" (distinguishing between the version of a gene you got from your mother vs. your father).
- **No GC Bias:** Unlike Illumina, PacBio does not require PCR amplification during library prep, meaning it sequences "hard-to-reach" AT-rich and GC-rich regions with uniform coverage.
- **Kinetics & Methylation:** Like Nanopore, PacBio can detect DNA methylation by measuring the slight "pulse" delay in the enzyme's speed as it hits a modified base.

Disadvantages of PacBio

- **Lower Throughput:** Compared to a NovaSeq (Illumina), PacBio produces much less data per run. You need many more SMRT cells to sequence a human genome to the same depth.
- **High Cost per Gigabase:** Due to the lower throughput and expensive consumables, PacBio is generally the most expensive of the three major platforms per unit of data.
- **Sample Requirements:** To get 20\$ kb HiFi reads, you must start with very high-quality, "unfragmented" DNA. If your DNA is slightly degraded (e.g., from old clinical samples), the technology won't work effectively.
- **Large File Sizes:** Because the machine literally records a "movie" of the light pulses, the initial raw data files are massive and require significant computing power to process into final HiFi reads.

Comparison of Sequencing Technologies

Feature	Illumina	Oxford Nanopore	PacBio (HiFi)
Read length	Short (150 - 300 bp)	Ultra-long (10 kb - 1 Mb)	Long (15 - 20 Kb)
Accuracy	High (99.9%)	Moderate (95 - 99%)	High (99.9%)
Primary Use	Identifying SNPs, clinical	Field work, structural variants	<i>De novo</i> assembly, phasing

Common File Formats

Common File Formats

- There are a lot of standard bioinformatic file formats
- We will introduce the ones you're most likely to come across
- Bioinformatics files generally come in two types:
 - **Binary** - store data more efficiently; **NOT** human readable
 - **Text** - easy to access, file sizes can be very large unless compressed

CSV (Comma-Separated Values)

- **Delimiter:** A comma (,).
- **Usage:** The most common format for general data exchange.
- If your data contains commas (e.g., "Cambridge, UK"), the file structure can break unless the text is wrapped in double quotes. This makes CSVs slightly more prone to parsing errors.

```
sample,fastq_1  
sample01,sample1_fastq.gz
```

TSV (Tab-Separated Values)

- **Delimiter:** A horizontal tab (`\t`).
- **Usage:** The gold standard for bioinformatics pipelines (e.g., BLAST outputs, VCFs).
- **The Advantage:** Since tabs almost never appear in biological names or DNA sequences, TSV files are much more robust and easier for programming languages to read.

```
sample    fastq_1  
sample01  sample1_fastq.gz
```

FASTQ

FASTQ files store both the **nucleotide sequence** and a **quality score** for every single base called by the sequencer.

```
1 @SEQ_ID:1:1101:1000:2000#0/1
2 GATTTGGGGTTCAAAGCAGTATCGATCAAATAGTAAATCCATTTGTTCAACTCACAGTTT
3 +
4 !''*(((((***+))%%%++) (%%%)) .1***-+*'')) **55CCF>>>>>CCCCCCC65
```

Line	Name	Description
1	Sequence ID	Starts with an @ character. It contains the unique ID of the sequencing read
2	Sequence	The actual string of nucleotides (A, C, G, T, N) as called by the machine
3	Separator	Starts with a + sign. Usually left blank to save space
4	Quality scores	String of ASCII characters. Each character represents a numerical Phred Score

PHRED Scores

- A **Phred quality score (Q)** is assigned to each nucleotide base call in a FASTQ file.
- It tells us how much we can trust the sequencer's decision for that specific base.
- It is a logarithmic measure of the probability that a base call is incorrect (**P**).
- The relationship between the score and the error probability is defined by the formula:

$$Q = -10 \log_{10} P$$

- Small increases in **Q** represent massive improvements in accuracy

PHRED Scores

PHRED score (Q)	P	Base call accuracy
10	1 in 10	90%
20	1 in 100	99%
30	1 in 1000	99.9%
40	1 in 10000	99.9%

FASTA

- The FASTA format is used for representing either nucleotide sequences (DNA/RNA) or peptide sequences (proteins).
- Unlike FASTQ, it **does not** contain quality scores

```
>NM_000546.6 Homo sapiens tumor protein p53 (TP53), mRNA
GTGAGCGCTTCGAGATGTTCCGAGAGCTGAATGAGGCCTTGGAAGCTCAAGGATGCCAGGCTGGGAAGGA
GCCAGGGGGGAGCAGGGCTCACTCCAGCCACCTGAAGTCCAAAAAGGGTCAGTCTACCTCCCGCCATAAA
AAACTCATGTTCAAGACAGAAGGGCCTGACTCAGACTGA
```

Component	Description
Header line	Starts with a greater-than symbol (>).Contains a unique identifier (e.g., an accession number like NC_000546.6).
Sequence	The actual letters (A, C, G, T, N for DNA or single-letter codes for Amino Acids)

SAM/BAM

- The **SAM (Sequence Alignment Map)** format is a TAB-delimited text file that
- Contains sequencing reads that have been aligned (mapped) to a reference sequence
- It is the standard output for aligners like BWA, Bowtie2, and STAR.
- **BAM** are compressed SAM files in binary format (much smaller file size).

```
@HD VN:1.6 SO:coordinate
@SQ SN:chr22 LN:50818468
@PG ID:bwa PN:bwa VN:0.7.17-r1188 CL:bwa mem ref.fa r1.fq r2.fq
read_01 99 chr22 10500 60 150M = 10650 300 GATCT... BCFFF... NM:i:0
read_02 16 chr22 20100 30 100M2I48M * 0 0 TTAGC... AABFF... AS:i:140
```

SAM/BAM

```
@HD VN:1.6 SO:coordinate
@SQ SN:chr22 LN:50818468
@PG ID:bwa PN:bwa VN:0.7.17-r1188 CL:bwa mem ref.fa r1.fq r2.fq
read_01 99 chr22 10500 60 150M = 10650 300 GATCT... BCFFF... NM:i:0
read_02 16 chr22 20100 30 100M2I48M * 0 0 TTAGC... AABFF... AS:i:140
```

Line(s)	Description
@HD	File metadata (version, sorting order)
@SQ	Reference sequence names (e.g., chr22) and their lengths
@PG	The specific command line used to generate the file (provenance).
Alignment	Each line represents a single read and its relationship to the reference genome. It consists of 11 mandatory fields .

SAM/BAM

```
read_01    99    chr22    10500    60    150M =    10650    300    GATCT...    BCFFF...    NM:i:0
read_01    147    chr22    10650    60    150M =    10500    -300    CCGAT...    BBCFF...    NM:i:0
read_02    16     chr22    20100    30    100M2I48M *    0    0    TTAGC...    AABFF...    AS:i:140
```

Field	Name	Description
1	QNAME	Query template NAME (The ID from the FASTQ file).
2	FLAG	A bitwise integer describing the read
3	RNAME	Reference sequence NAME (chr22)
4	POS	The 1-based leftmost mapping POSition.
5	MAPQ	MAPping Quality (Phred scale)
6	CIGAR	Compact Idiosyncratic Gapped Alignment Report

CIGAR strings

read_02 16 chr22 20100 30 **100M2I48M** * 0 0 TTAGC... AABFF... AS:i:140

Code	Operation	Description
M	Alignment match	Base aligns to the reference (can be a sequence match or a mismatch)
I	Insertion	Extra bases in the read that are not present in the reference genome.
D	Deletion	Bases present in the reference but missing from the read (a gap).
S	Soft clipping	Bases at the ends of the read that did not align and are ignored by the aligner.
H	Hard clipping	Similar to soft clipping, but the clipped bases are physically removed from the SEQ field.
N	Skipped regions	Used in RNA-seq to represent introns (gaps between exons).

CIGAR strings

```
read_02    16    chr22    20100    30    100M2I48M *    0    0    TTAGC...    AABFF...    AS:i:140
```

Decoding 100M2I48M

Imagine a 150bp read with this CIGAR string:

1. **100M**: The first 100 bases align to the reference genome.
2. **2I**: The next 2 bases in the read are an **insertion** (they don't exist in the reference).
3. **48M**: The final 48 bases align to the reference.

VCF

- The **VCF (Variant Call Format)** is the standard text-based format for storing gene sequence variations.
- After we align our reads (BAM) and run a "Variant Caller" (like GATK), the output is a VCF file.

```
##fileformat=VCFv4.2
##FILTER=<ID=PASS,Description="All filters passed">
##reference=file:///seq/references/Homo_sapiens_assembly38.fasta
##INFO=<ID=DP,Number=1,Type=Integer,Description="Total Depth">
##INFO=<ID=AF,Number=A,Type=Float,Description="Allele Frequency">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
#CHROM    POS    ID    REF  ALT  QUAL FILTER    INFO FORMAT    HG001
chr17     7673776    rs12345    A    G    100    PASS DP=25;AF=0.5    GT:GQ    0/1:99
chr17     7673802    .    C    CT    80    PASS DP=22;AF=1.0    GT:GQ    1/1:85
```

VCF

```
##fileformat=VCFv4.2
##FILTER=<ID=PASS,Description="All filters passed">
##reference=file:///seq/references/Homo_sapiens_assembly38.fasta
##INFO=<ID=DP,Number=1,Type=Integer,Description="Total Depth">
##INFO=<ID=AF,Number=A,Type=Float,Description="Allele Frequency">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
#CHROM    POS    ID      REF     ALT    QUAL    FILTER  INFO    FORMAT  HG001
chr17     7673776    rs12345    A       G      100     PASS    DP=25;AF=0.5    GT:GQ      0/1:99
chr17     7673802    .         C       CT     80      PASS    DP=22;AF=1.0    GT:GQ      1/1:85
```

Component	Description
Metadata lines	Lines starting with ## . These define the reference genome used, the software that called the variants, and explanations for the data columns (INFO and FORMAT).
Header line	A single line starting with #CHROM that labels the columns.
Data lines	Each line represents a single genetic variant (a SNP, Indel, or Structural Variant).

CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO	FORMAT	HG001
chr17	7673776	rs12345	A	G	100	PASS	DP=25;AF=0.5	GT:GQ	0/1:99
chr17	7673802	.	C	CT	80	PASS	DP=22;AF=1.0	GT:GQ	1/1:85

Column	Description
CHROM	The chromosome where the variant is located
POS	The 1-based position on the chromosome
ID	Identifier (e.g., a dbSNP "rs" number)
REF	The reference allele (what is in the reference genome)
ALT	The alternative allele (what was found in the sample)
QUAL	Phred-scaled quality score for the assertion of a variant
FILTER	PASS if it met all filters; otherwise, the reason for failure.
INFO	Additional metadata (e.g., depth of coverage, allele frequency)

BED

- The **BED (Browser Extensible Data)** format is a lightweight, tab-delimited text file used to define specific **regions** or "features" on a genome.
- While a VCF focuses on *variants* (specific changes), a BED file is typically used to define areas of interest, such as:
 - **Exons** of a gene.
 - **ChIP-seq peaks** (where a protein binds).
 - **Target regions** for exome sequencing (the "Capture Bait" areas).

chr7	127471196	127472363	Pos_Control	1000	+
chr7	127473530	127474214	Neg_Control	500	-
chr8	1523400	1523500	Peak_Region	900	+

- The **GFF (General Feature Format)** is a standard, tab-delimited text file used for describing genomic features such as genes, exons, transcripts, and regulatory elements.
- While a BED file is "lightweight" and simple, GFF (and its successor **GTF**) is highly structured and hierarchical, often used as the "Annotation" file to tell tools where genes are located.

```
chr19  RefSeq  gene  58858172  58874214  .  -  .  ID=gene:A1BG;Name=A1BG
chr19  RefSeq  mRNA  58858172  58874214  .  -  .  ID=rna:NM_130786.4;Parent=gene:A1BG
```

```
chr19 RefSeq gene 58858172 58874214 . - . ID=gene:A1BG;Name=A1BG
chr19 RefSeq mRNA 58858172 58874214 . - . ID=rna:NM_130786.4;Parent=gene:A1BG
```

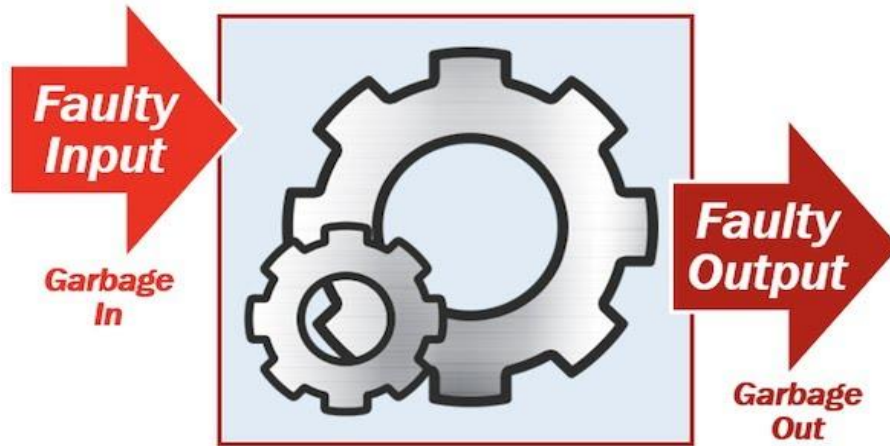
Column	Name	Description
1	seqid	The chromosome or scaffold name
2	source	The software or database that generated the feature
3	type	The type of feature (Gene, mRNA, Exon, CDS)
4	start	1-based start position of the feature
5	end	1-based end position (inclusive)
6	score	A floating-point value (or . if not available)
7	strand	Forward (+) or Reverse (-)
8	phase	For CDS features, indicates the reading frame (0, 1, or 2)
9	attributes	A semicolon-separated list of tag-value pairs

Summary

- CSV / TSV → comma-separated or tab-separated files storing tabular data, common output for many analysis software
- FASTQ → store (“raw”) sequencing data and quality scores for Illumina and base called ONT reads.
- FASTA → store DNA sequences, for example reference genomes or genome assemblies
- SAM/BAM → store alignment data, including sequence and quality information from the FASTQ file
- VCF → store genetic variance information based on the position of the reference genome
- BED/GFF → tab-delimited file formats that store the coordinates of genomic features, such as gene annotations for a genome or primer locations.

Sequence Quality Control (QC)

Introduction to Sequence Quality Control



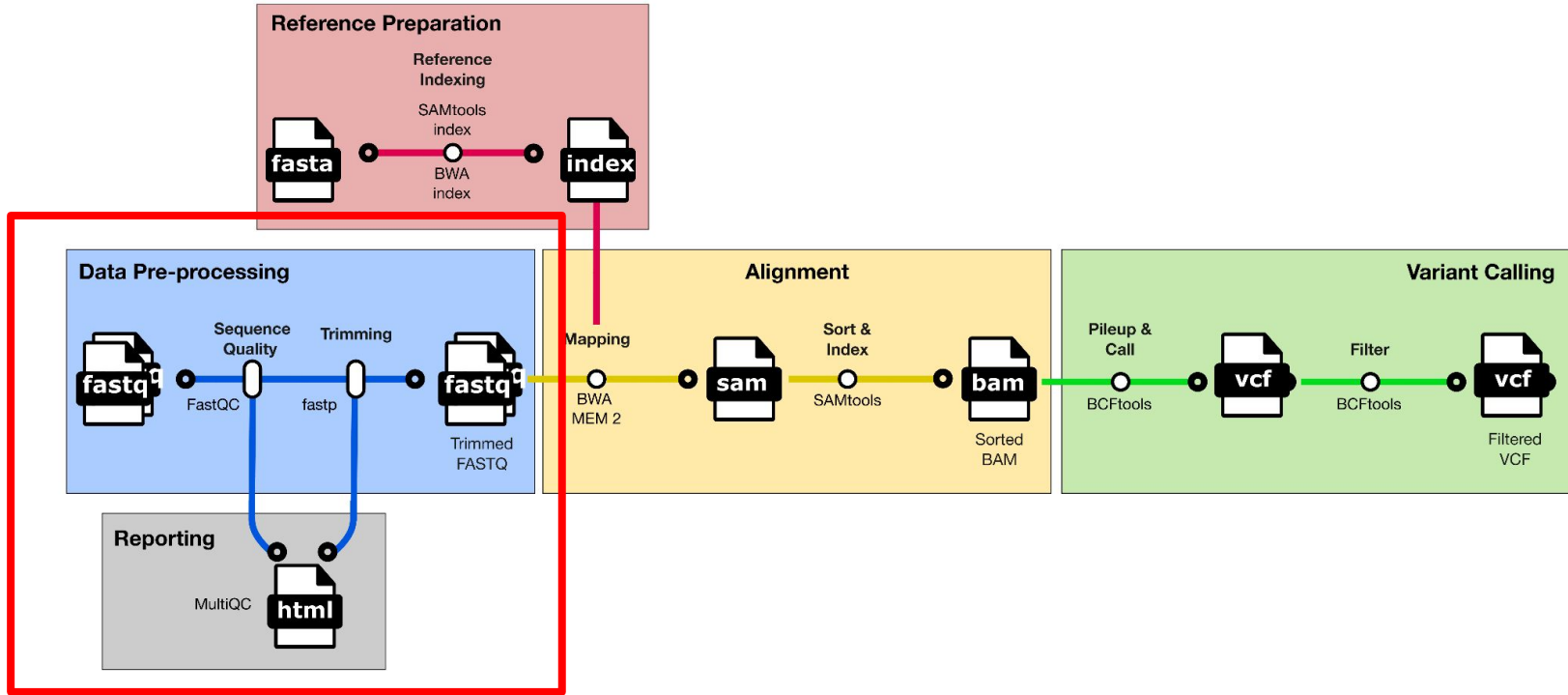
- It is standard practice to perform **Quality Control (QC)** on the raw sequencing reads before any actual analysis
- It is the process of evaluating the quality of reads in each sample
- Biologically meaningful results from NGS data depend on the quality of the raw reads.

Introduction to Sequence Quality Control

Typically we would perform the following as part of our NGS QC:

- Check for primers and adaptors
- Trim low quality ends
- Remove low quality reads
- Remove short sequences
- Remove reads with ambiguous bases (“N”)
- Remove duplicates

Overview



FastQC

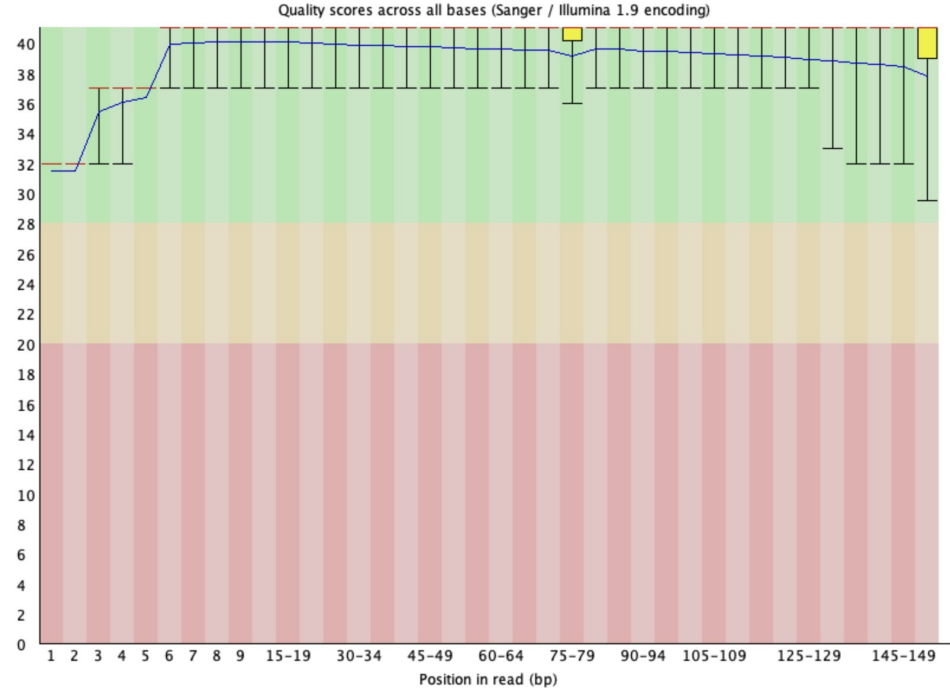
- We will use a tool called FastQC to check our data
- Most sequencers will generate a QC report as part of their analysis pipeline, but this is usually only focused on identifying problems which were generated by the sequencer itself.
- FastQC aims to provide a QC report which can spot problems which originate either in the sequencer or in the starting library material.

FastQC

Function	A quality control tool for high throughput sequence data.
Language	Java
Requirements	A suitable Java Runtime Environment The Picard BAM/SAM Libraries (included in download)
Code Maturity	Stable. Mature code, but feedback is appreciated.
Code Released	Yes, under GPL v3 or later .
Initial Contact	Simon Andrews
Download Now	

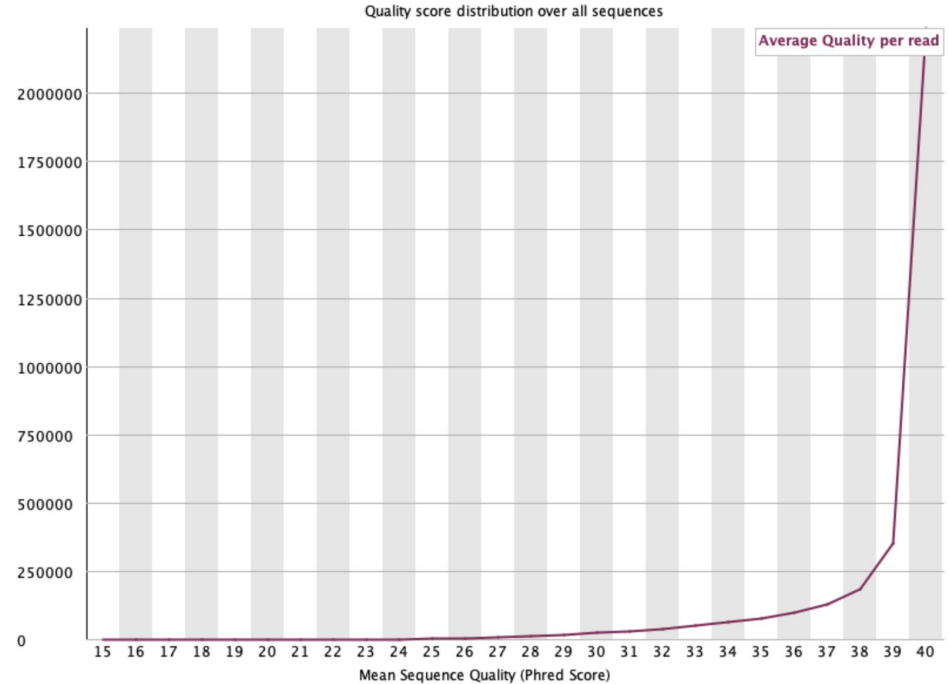
Per Base Sequence Quality

- Shows an overview of the range of quality values across all bases at each position in the FASTQ file
- The higher the score the better the base call
- The background of the graph divides the y axis into very good quality calls (green), calls of reasonable quality (orange), and calls of poor quality (red)
- it is common to see base calls falling into the orange area towards the end of a read



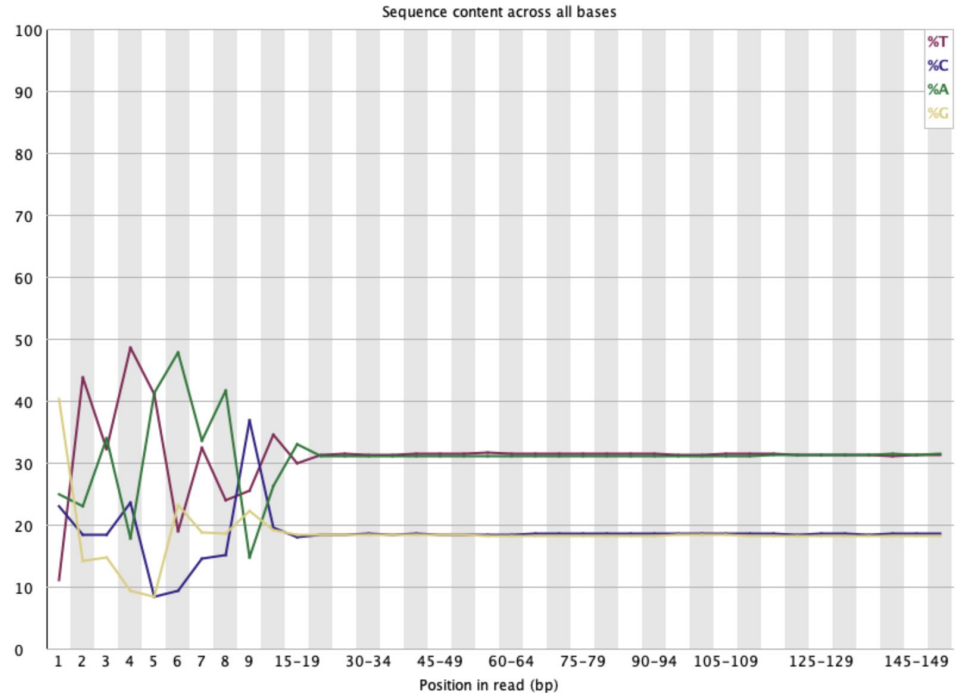
Per Sequence Quality

- Allows you to see if a subset of your sequences have universally low quality values
- By chance, some of your sequences will have low quality
- If a majority of your sequences are low quality ($Q < 30$), this suggests a systematic issue (e.g. end of a flowcell)



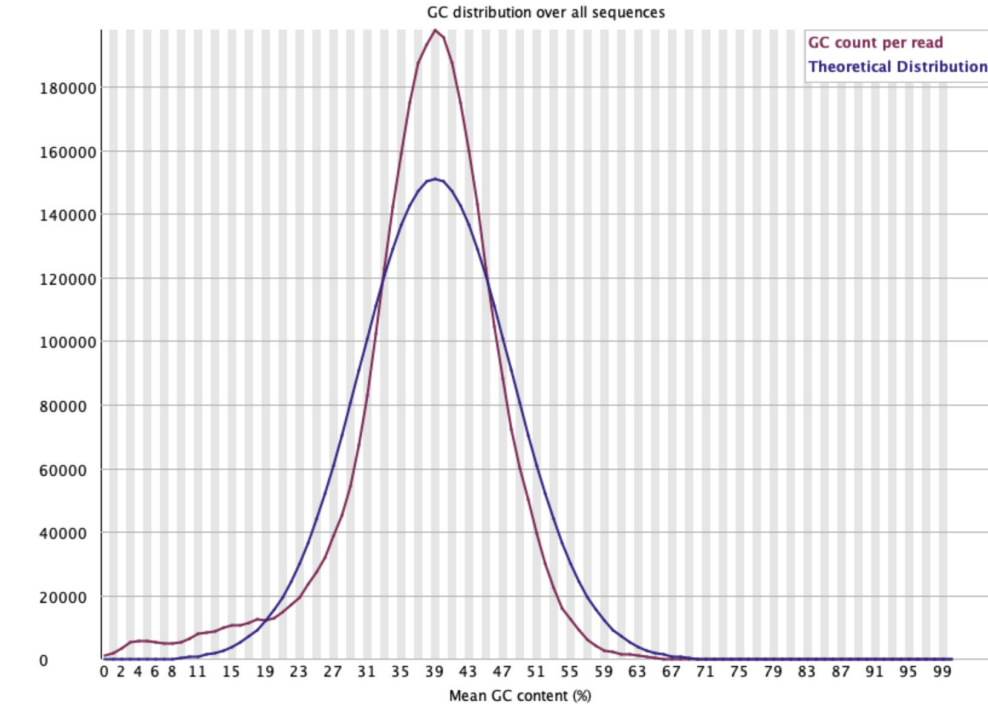
Per Base Sequence Content

- Shows the proportion of each base position in a file for which each of the four normal DNA bases has been called
- In a random library there should be little to no difference between the different bases of a sequence run, so the lines in this plot should run parallel with each other
- The relative amount of each base should reflect the overall amount of these bases in your genome



Per Sequence GC Content

- Measures the GC content across the whole length of each sequence in a file and compares it to a modelled normal distribution of GC content
- You should see a roughly normal distribution of GC content where the central peak corresponds to the overall GC content of the underlying genome
- An unusually shaped distribution could indicate a contaminated library or some other kinds of biased subset of your data

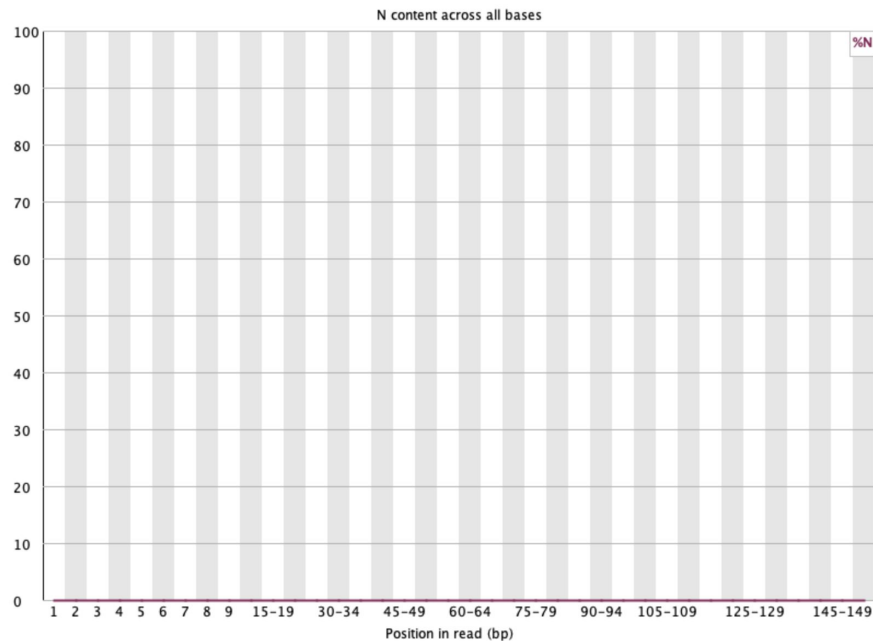


Per Base N Content

- If a sequencer is unable to make a base call with sufficient confidence then it will normally substitute an N rather than a conventional base call
- This plot shows the percentage of base calls at each position for which an N was called
- It's common to see a very low proportion of Ns appearing in a sequence, especially nearer the end of a sequence
- A higher proportion suggests the sequencer had a problem making the base calls

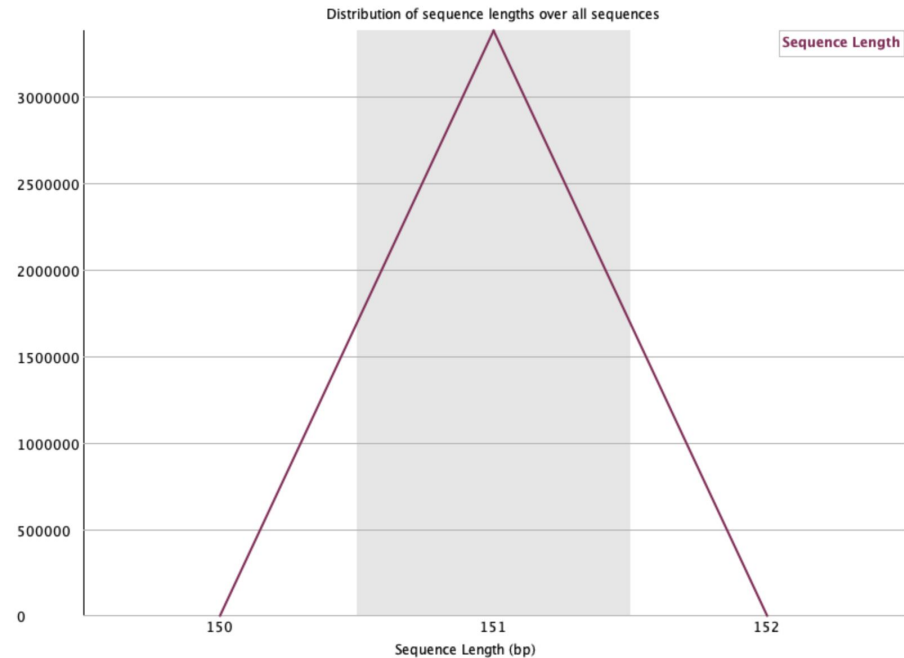


Per base N content



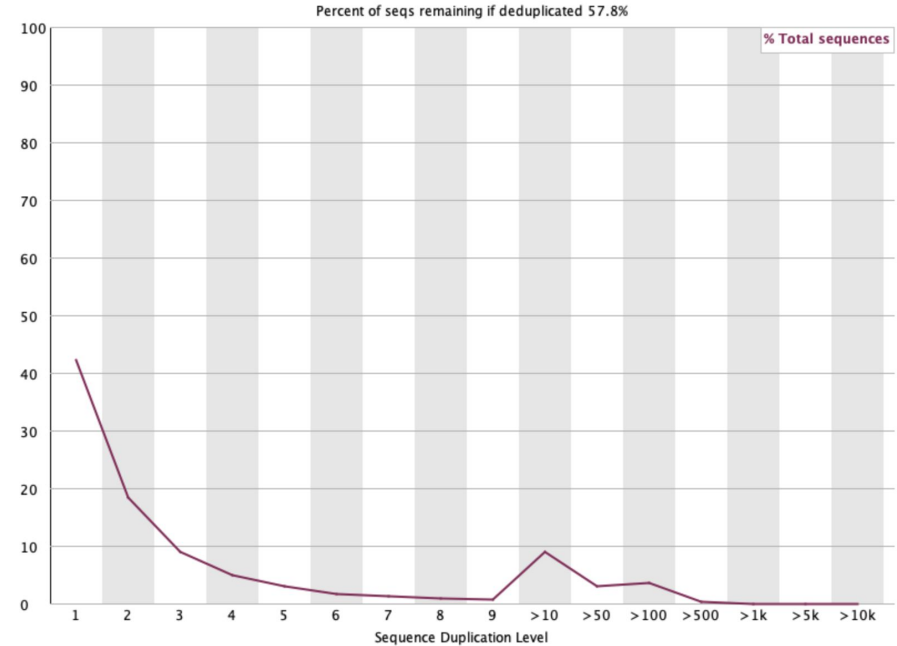
Sequence Length Distribution

- This plot shows the distribution of fragment sizes in the file which was analysed
- For Illumina which generates sequence fragments of uniform length, there should only be one peak at the expected fragment size (e.g 151 bp)
- For variable length FastQ files this will show the relative amounts of each different size of sequence fragment (ONT/PacBio)



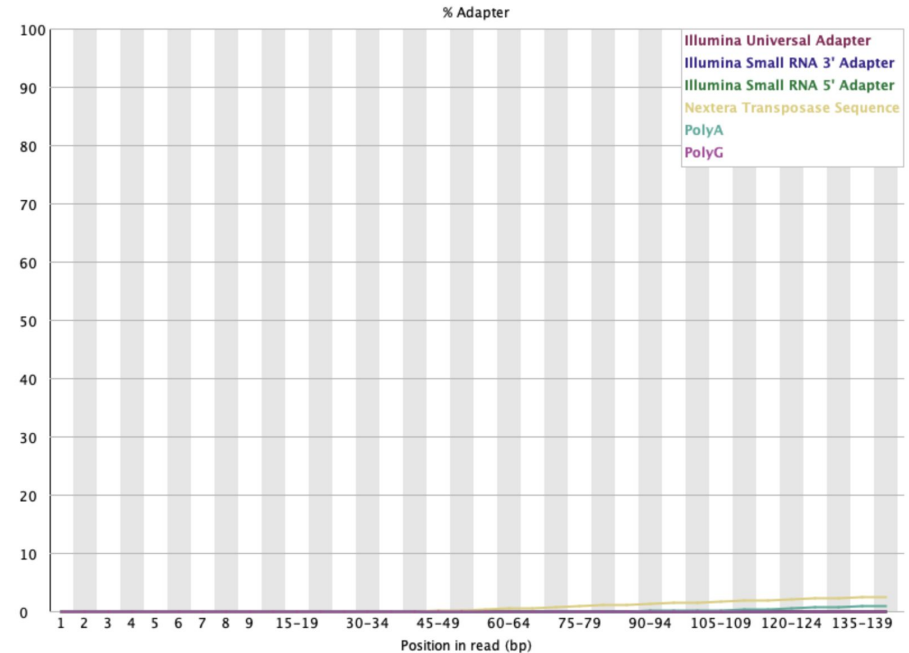
Sequence Duplication Levels

- fastQC counts the degree of duplication for every sequence in a library and creates a plot showing the relative number of sequences with different degrees of duplication
- In a diverse library most sequences will occur only once in the final set. A low level of duplication may indicate a very high level of coverage of the target sequence, but a high level of duplication is more likely to indicate some kind of enrichment bias (eg PCR over amplification)



Adapter Content

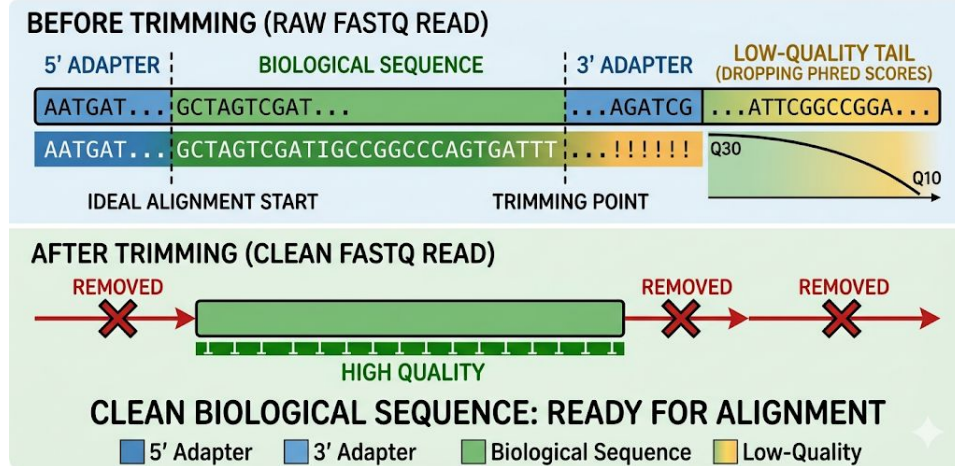
- This plot shows a cumulative percentage count of the proportion of your library which has seen each of the adapter sequences at each position
- Once a sequence has been seen in a read it is counted as being present right through to the end of the read so the percentages you see will only increase as the read length goes on
- The percentage of adapter sequences should be less than 5%



Filtering and Trimming

- Raw NGS data contains technical artifacts, sequencing errors, and low-quality bases that can mislead alignment algorithms and variant callers.
- Filtering and trimming are the "sanitization" steps that ensure your downstream analysis is based on high-quality biological data.

WHAT DO WE MEAN BY TRIMMING READS?



- We will use a tool called fastp to filter and trim our sequence data
- Automatically clips low-quality bases from the 5' and 3' ends
- Automatically detects adapter sequences (and trims them via overlap analysis)
- Generates interactive HTML reports (similar to FastQC) and machine-readable JSON files

OpenGene/**fastp**

An ultra-fast all-in-one FASTQ preprocessor
(QC/adapters/trimming/filtering/splitting/merging...)



36

Contributors

376

Issues

2k

Stars

374

Forks



MultiQC

- **MultiQC** is a reporting tool that searches a given directory for analysis logs from various bioinformatics tools and compiles them into a **single, interactive HTML report**
- **Aggregate Statistics:** View a single table showing the total reads, GC content, and duplication rates for every sample in your project.
- **Interactive Plots:** Hover over lines to identify specific samples, zoom into regions of interest, and toggle specific samples on or off.
- **Tool Agnostic:** It supports over 100 different bioinformatics tools. If your pipeline uses BWA for alignment and GATK for variants, MultiQC will catch the logs from both.



MultiQC Summary Table

General Statistics

Copy table

Configure columns

Scatter plot

Violin plot

Export as CSV...

Showing 3/3 rows and 8/15 columns.

Summarize table

Sample Name	% Duplication	Reads After Filtering	GC content	% PF	% Adapter	Dups	GC	Seqs
SRR36763845	9.0 %	6.5 M	37.1 %	96.3 %	3.0 %			
SRR36763845_1						42.2 %	37.0 %	3.4 M
SRR36763845_2						34.4 %	37.0 %	3.4 M

- This is a compilation of statistics collected from the outputs of **fastp** and **fastQC**.
- Sequencing metrics such as the % of duplicated reads and GC content of the reads are shown
- This is a useful way of quickly identifying samples that are of lower quality due to poor sequencing

fastp filtered reads

Filtered Reads

Filtering statistics of sampled reads.

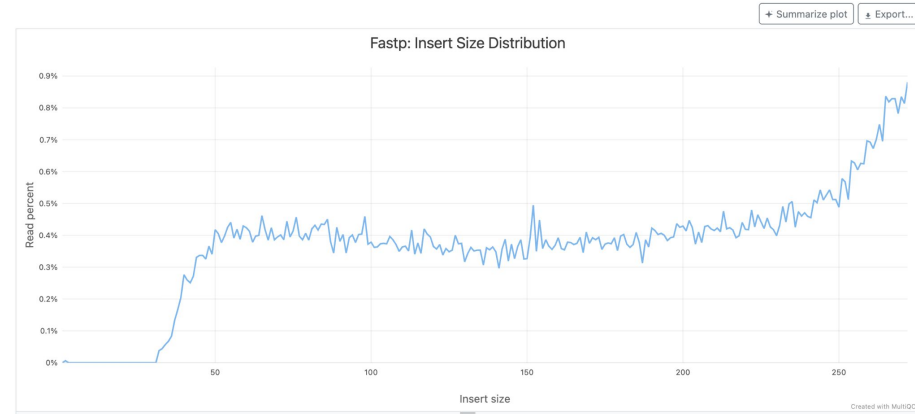


- This plot shows the results of the **fastp** read filtering step where reads are trimmed, adapters removed and low quality reads are thrown out.
- The reads that passed this step are highlighted in blue

fastp Insert size

- This plot shows the distribution of insert sizes for each set of sequence files
- There should be a peak around the insert size you're expecting

Insert size estimation of sampled reads.

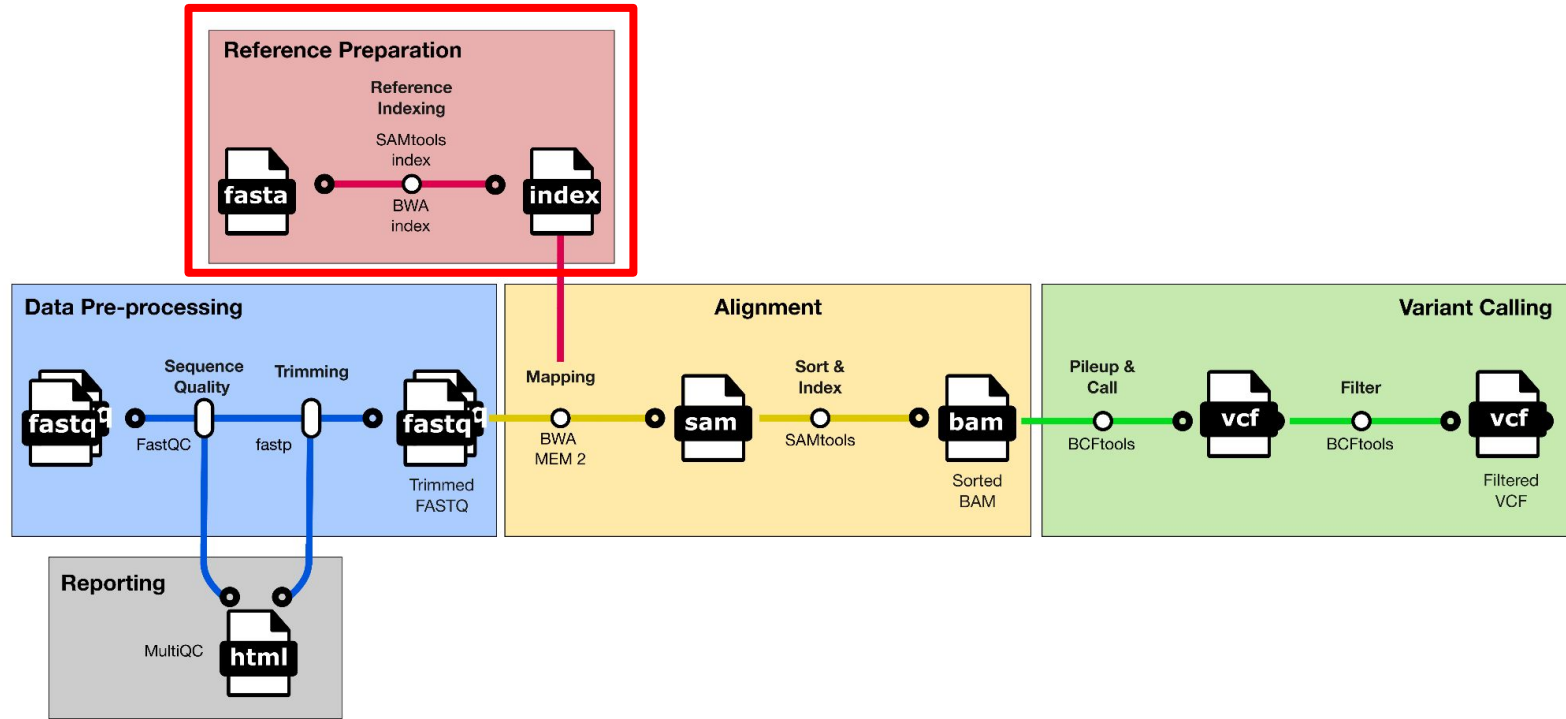


Practical & exercises

- Work through Chapter 5 and complete the exercise
- Work through Chapter 6 and complete the exercise

Reference Genomes

Overview



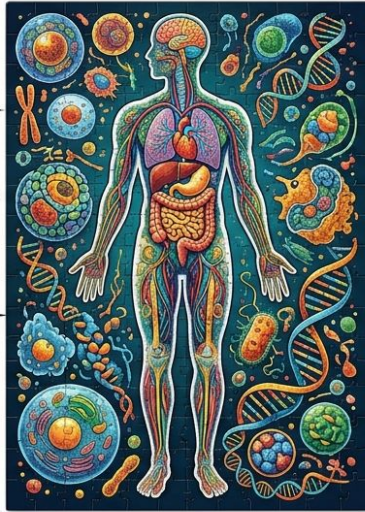
Reconstructing our genomes

THE GENOME ASSEMBLY PUZZLE ANALOGY

THE REFERENCE GENOME
(A Finished Puzzle)


REFERENCE CHROMOSOMES


KNOWN SEQUENCE



RAW SEQUENCING READS
(Shredded Pieces)

**DNA
FRAGMENTATION
& SEQUENCING**



SHORT READ
(ILLUMINA)

LONG READ
(ONT/PACBIO)

DE NOVO ASSEMBLY
(No Reference)



CONTIGS

Harder with short reads,
easier with long reads

REFERENCE MAPPING
(With Guide)



Fast and efficient for
known genomes

COMPLEX REGIONS
(Repeats)



Short reads fail in highly
repetitive blue sky; Long
reads span the repeat

Reference Genomes

- A reference genome is a high-quality, representative sequence of the complete set of DNA for a specific species (the “picture” of the puzzle).
- It is typically stored in FASTA format and serves as a gold standard.
- Selecting a suitable reference sequence is an **important step**.
- If your reference is too divergent from your sample it can affect read mapping and possibly the consensus sequence
- Make sure you select the right species!

HOW DO I FIND MY REFERENCE GENOME?

STEP 1: DEFINE YOUR SAMPLE

? SPECIES IDENTIFICATION



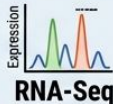
Is it a model organism?
(Human, Mouse, *E. coli*)

BIOLOGICAL SOURCE



Is it derived from a tissue sample,
cell line, or metagenomic population?

DOWNSTREAM ANALYSIS



Are you looking for SNPs, gene
expression, or structure?

GUIDING DECISIONS

STEP 2: SELECT THE DATABASE & BUILD

A. MODEL ORGANISMS (Best Annotation)

GRCh38
(Human)

UCSD Genome
Browser

mm10/GRCm39
(Mouse)

Ensembl

Use high-quality, community-standard builds

B. NON-MODEL / NOVEL SPECIES (Search for Best Fit)



NCBI
Assembly

RefSeq

Search for nearest-relative genome or
download *de novo* contigs

STEP 3: ACQUIRE THE FILES



**REFERENCE
FASTA**
(e.g., hg38.fa)



**ANNOTATION
GTF/GFF**
(e.g., hg38.gtf)

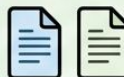


BWA



Bowtie2

**GENOME
INDEX**



.bwt, .rev,

⚠ CAUTION: Coordinate mismatches!
Ensure GFF/GTF annotation uses the same
assembly build as your FASTA reference.
Mixing 'hg19' and 'hg38' will ruin your analysis!

Reference Indexing

- Indexing is the process of creating a "table of contents" or a "search engine" for your reference genome.
- It allows alignment tools to jump instantly to the correct genomic coordinate.
- Every alignment tool (BWA, Bowtie2, STAR) requires its own specific index format. You must run the indexing command **once** before you can align any samples.
- While aligners need their own index to *map* reads, other downstream tools (like GATK or IGV) need a general-purpose index called a **FASTA Index**
- We can generate this index with the `samtools faidx` command

Practical & exercises

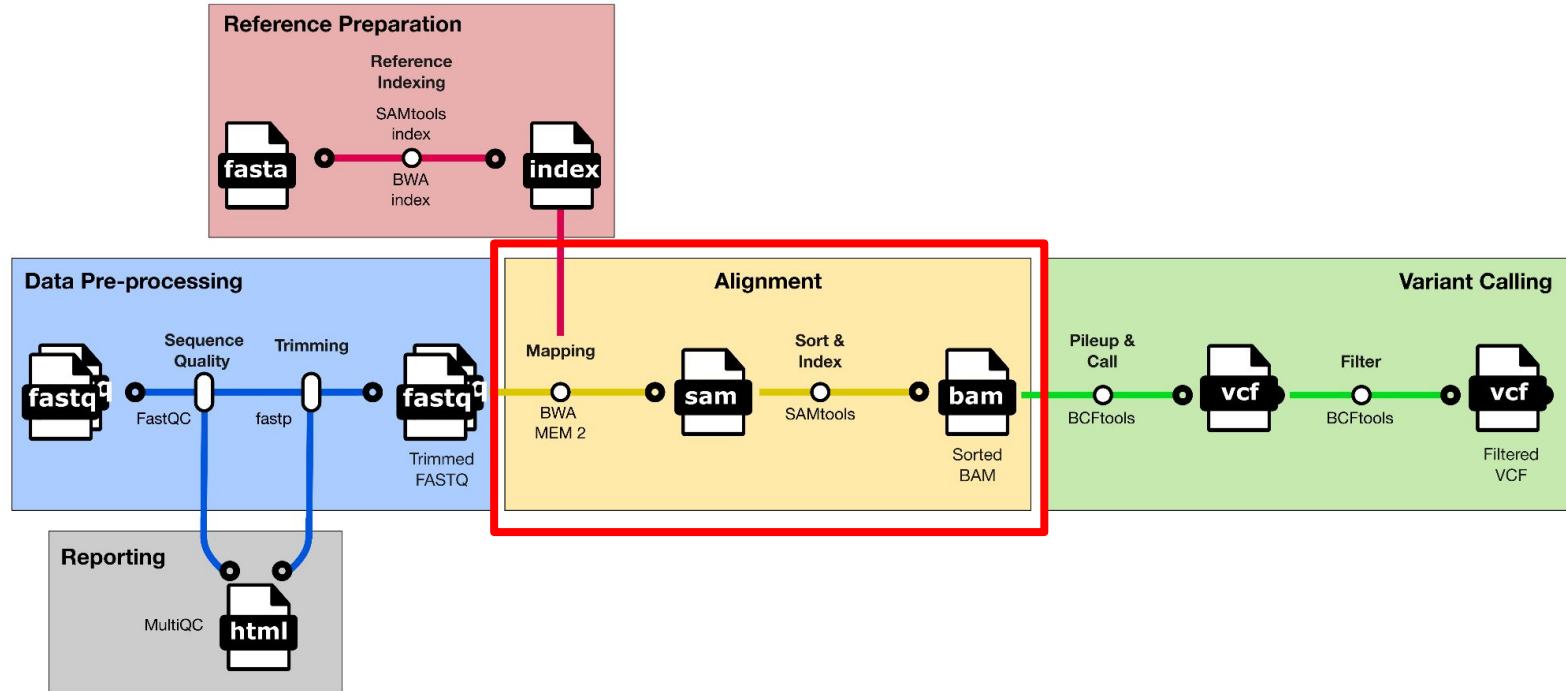
- Work through Chapter 7 and complete the exercises

Sequence Alignment (Mapping)

Sequence Alignment (Mapping)

- What is sequence alignment?
- How does it work?
- What tools can you use?
- What do the results look like?
 - Basic statistics
 - Coverage plots

Overview

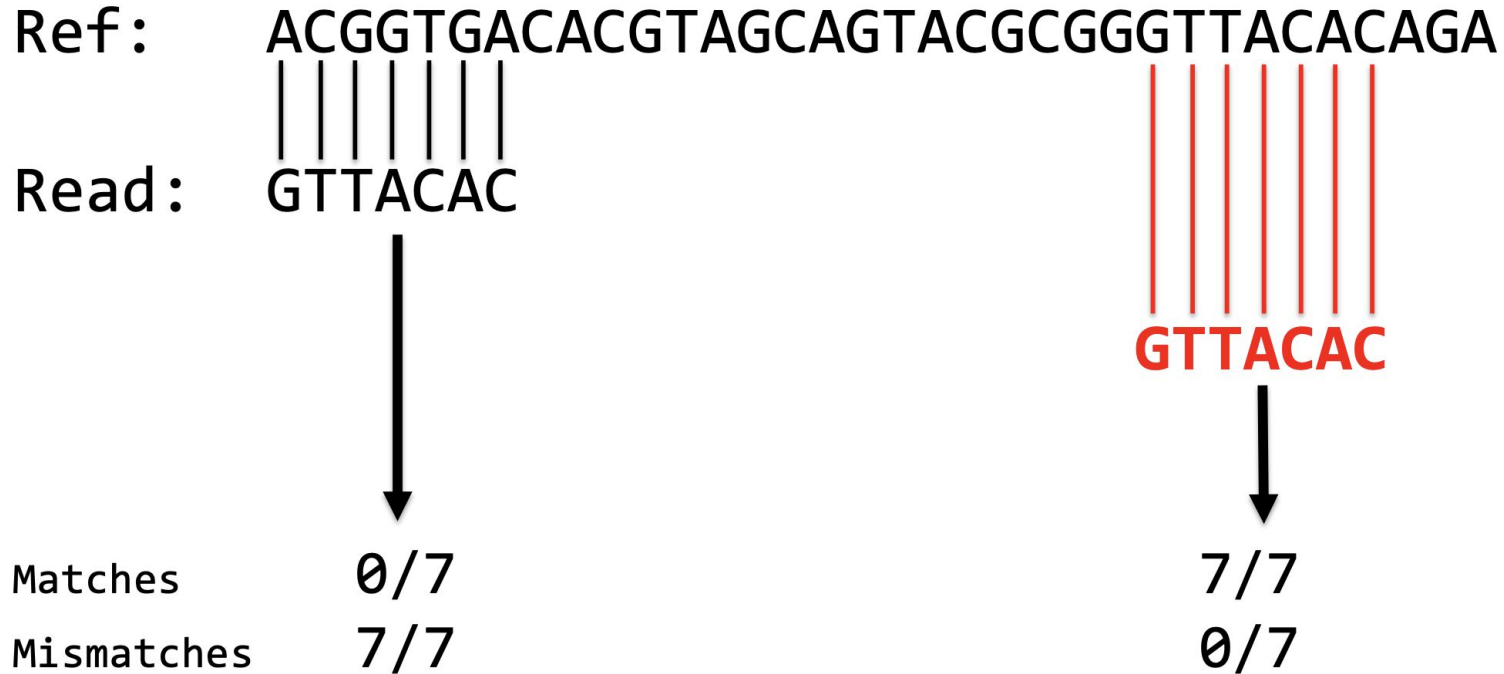


Aligning Reads to a Reference

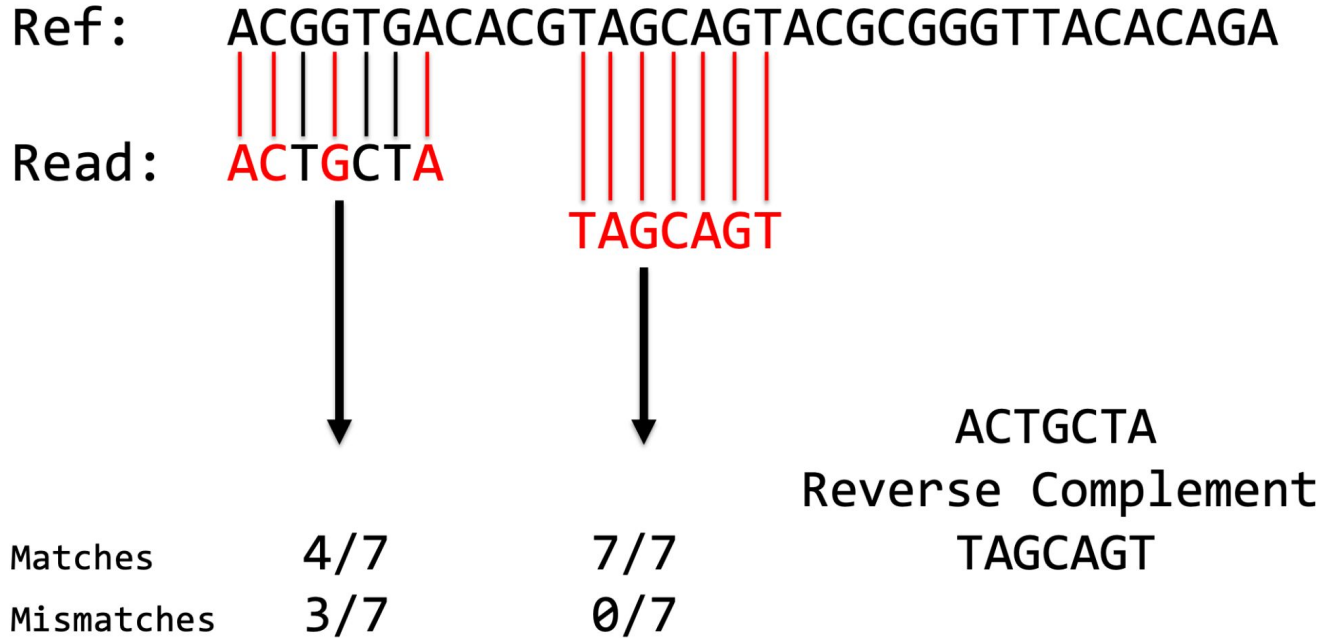
Ref: ACGGTGACACGTAGCAGTACGCGGGTTACACAGA

Read: GTTACAC

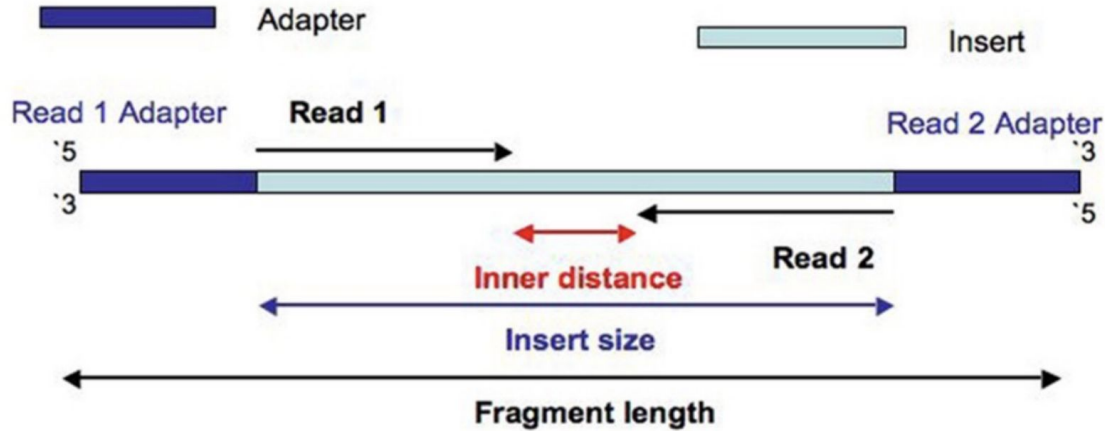
Aligning Reads to a Reference



Aligners check the reverse complement

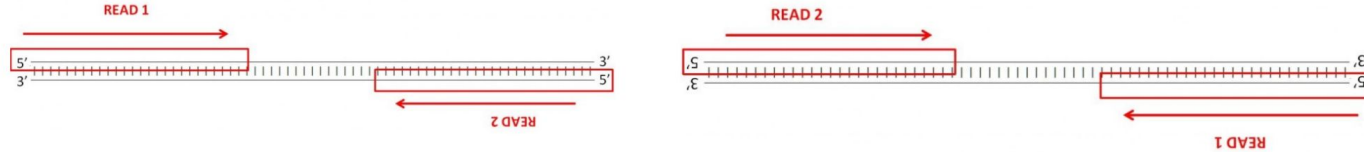


Paired end reads and insert size

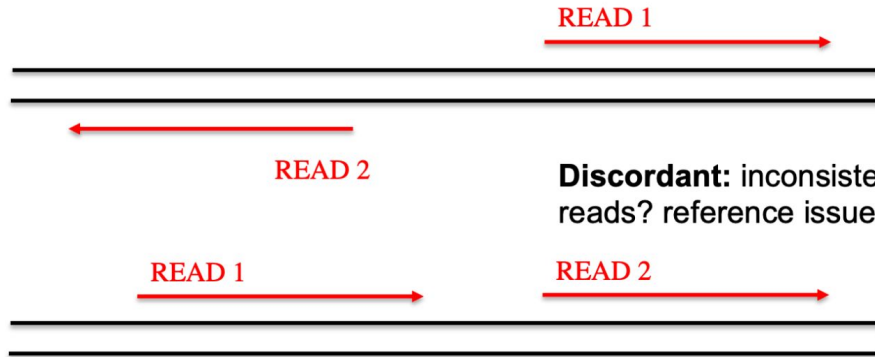


- Typically, the two reads do not overlap, but they can if the reads are long and fragments are short [redundant data, but can be used to correct errors]
- 500bp fragments + 2 x 300bp reads = 100bp overlap

Concordance and discordance of paired end reads



Concordant: consistent orientation of read pairs with respect to reference, have insert size within the expected range (depends on library)



Discordant: inconsistent orientation (mixed up reads? reference issues? Abnormal insert size)

Aligning Reads to a Reference: Mutations and Indels

Ref: ACGGTGACACGTAGCAGTACGCGGGTTACACAGA

ACGG**C**GA CAGT**T**CG AC-CAGA

 AG**A**CAGTA GCGGGTT

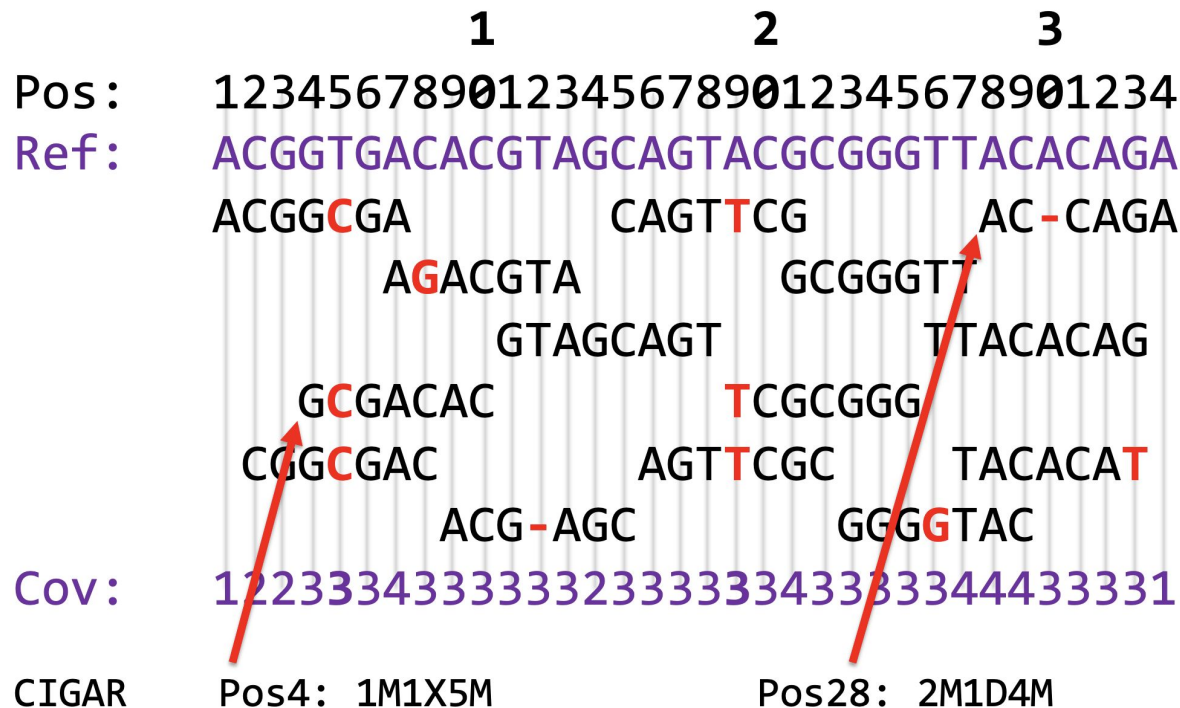
 GTAGCAGT TTACACAG

 G**C**GACAC **T**CGCGGG

 CGG**C**GAC AGT**T**CGC TACACAT**T**

 ACG-AGC GGG**G**TAC

CIGAR strings

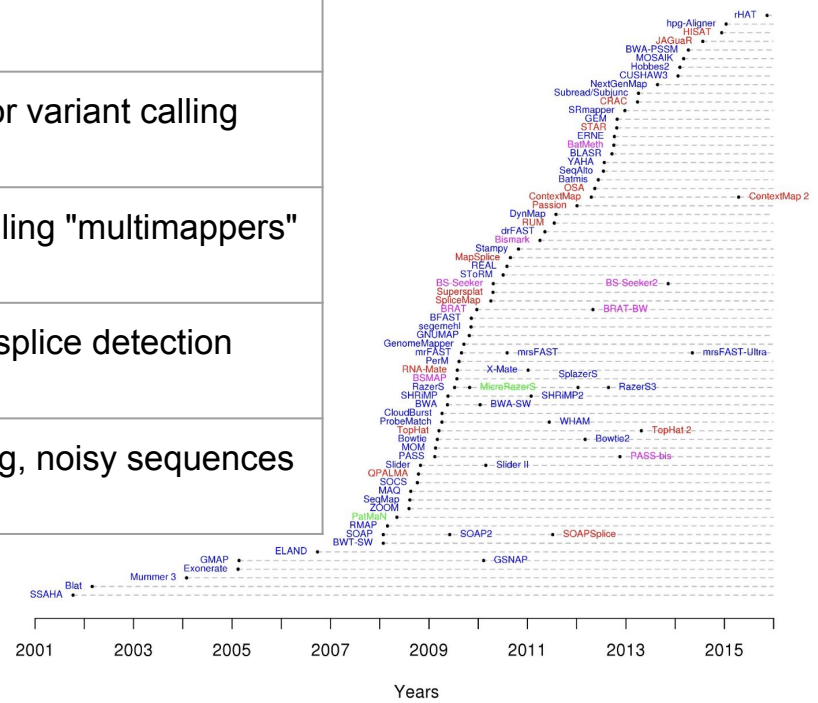


Unmapped Reads

- Reads that could not be aligned to the reference sequence are marked as unmapped
- What are these reads?
 - Host
 - Bacteria, parasites, viruses
 - Random "low complexity" sequences
- We are "targeting" a specific reference sequence to align against. This may not be the most appropriate reference for your species of interest

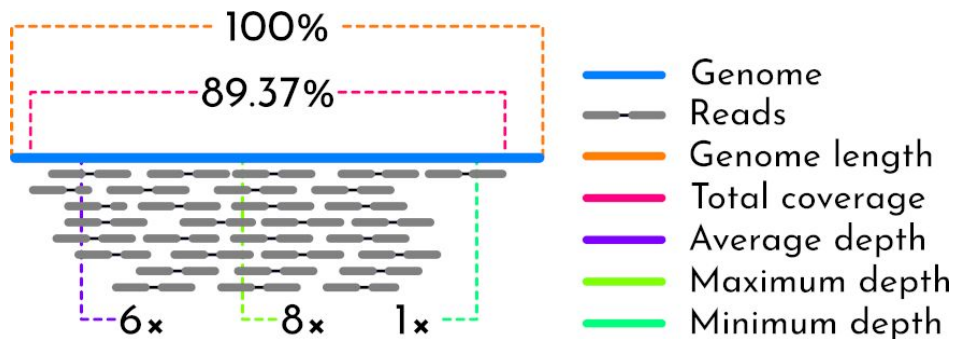
Which aligner should I use?

Application	Recommended tools	Key feature
WGS/WES	BWA-MEM 2	High sensitivity for variant calling
ChIP-Seq / ATAC-Seq	Bowtie-2	Excellent at handling "multimappers"
RNA-Seq	STAR	Best-in-class for splice detection
Long Reads	Minimap2	Optimized for long, noisy sequences



Terminology Alert

What does the word “**coverage**” mean?



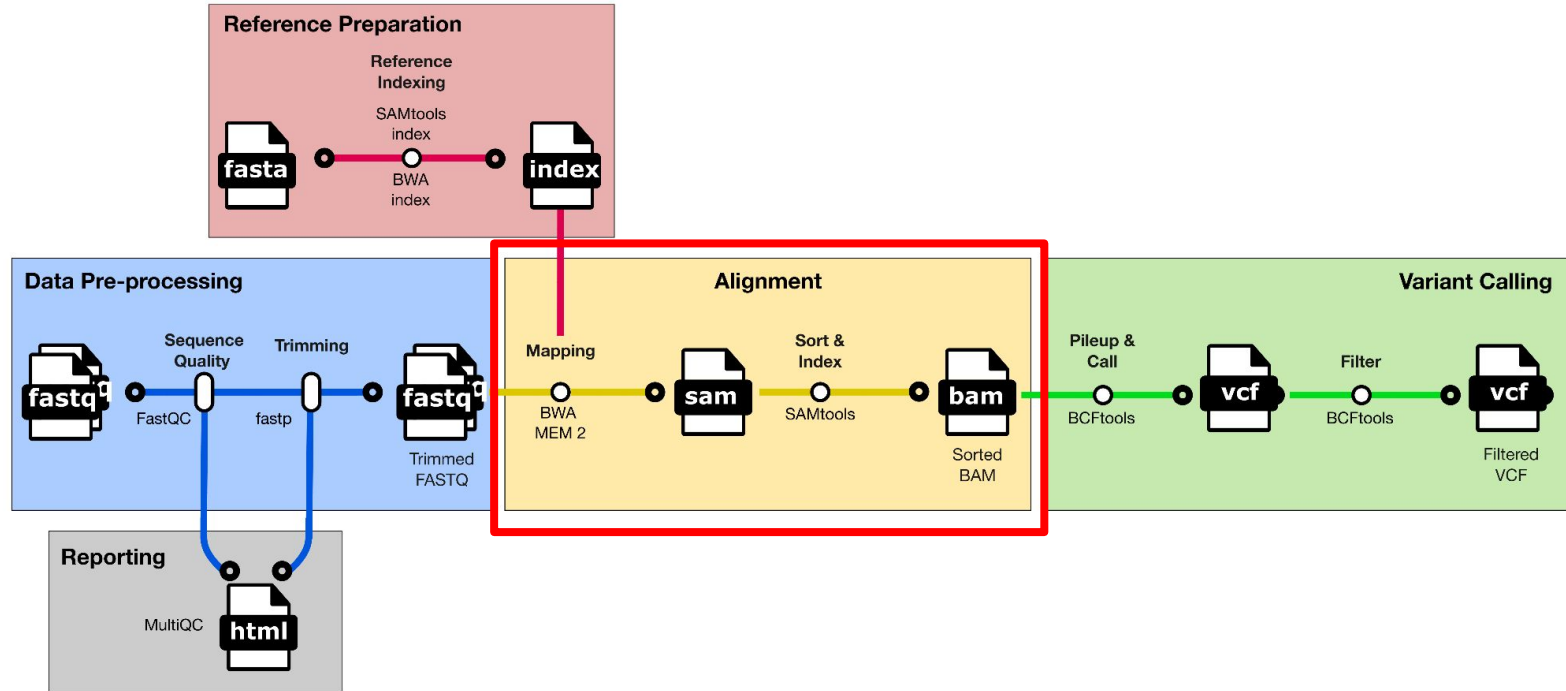
- **Depth of Coverage** - number of reads that align to a position/region of the genome

Example: a genome has 12x coverage, i.e. there is a median or mean of 12 reads aligned to that genome.

- **Sequence Coverage** - the fraction of a genome that has a sufficient number of reads for analysis

Example: My consensus genome has 82% coverage, i.e. we identified the sequence for 82% of the genome but have 18% with missing data.

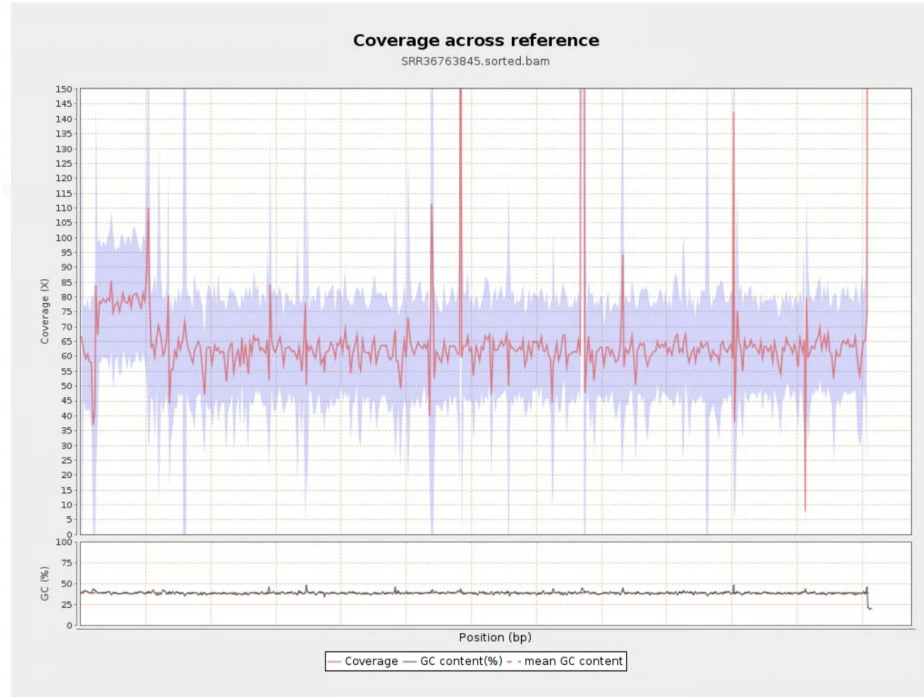
Overview



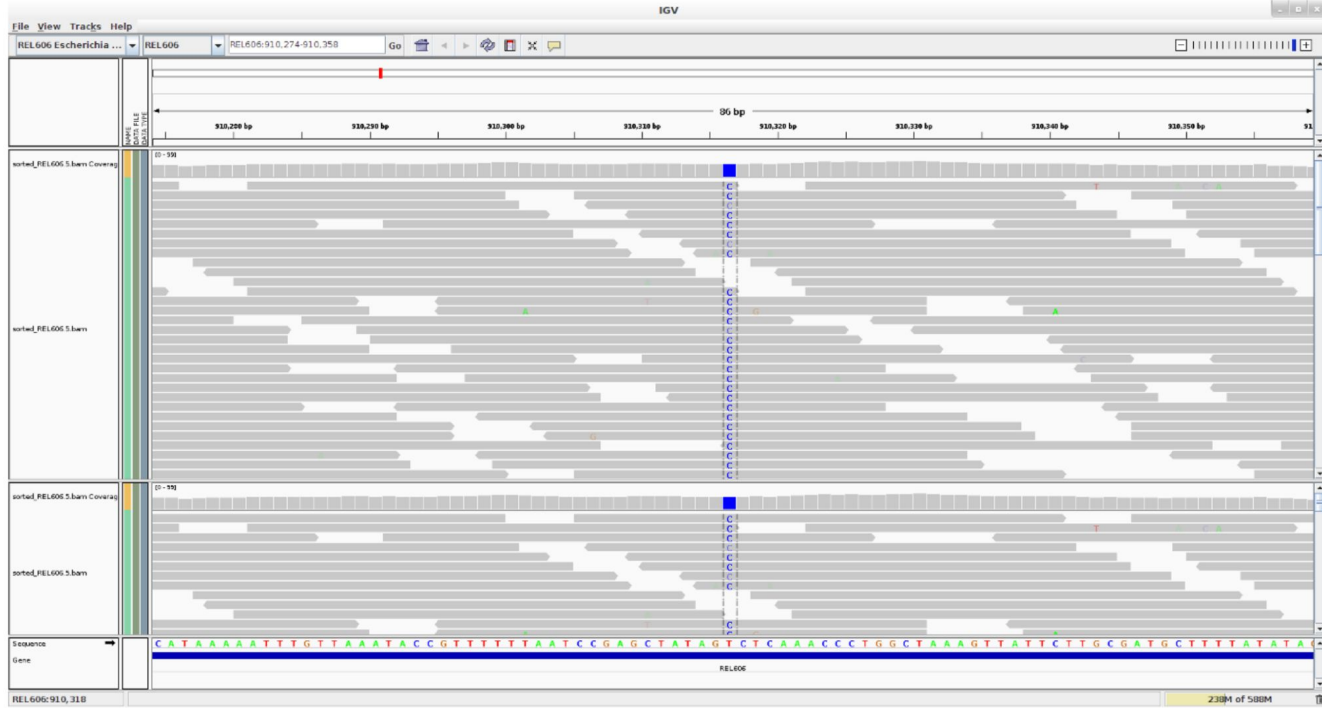
Overview

- Convert your **SAM** file into **BAM** files
- **BAM Binary Alignment/Map**
- BAM files are **compressed** binary versions of the same data (and **faster**)
- Initially the SAM/BAM is sorted by the order the reads were in their files
- **Sort** BAM file
 - All the reads where alignment starts at position 1 first
 - All the reads where alignment starts at position 2, then 3, then 4 etc
 - `samtools sort my.sam -o my.bam`
- **Index** the BAM file
 - Enables downstream tools (consensus/variant calling) to rapidly look up what is aligned to e.g. position 10,456
 - `samtools index my.bam`

Depth of Coverage



IGV (Mapping Visualisation - BAM)

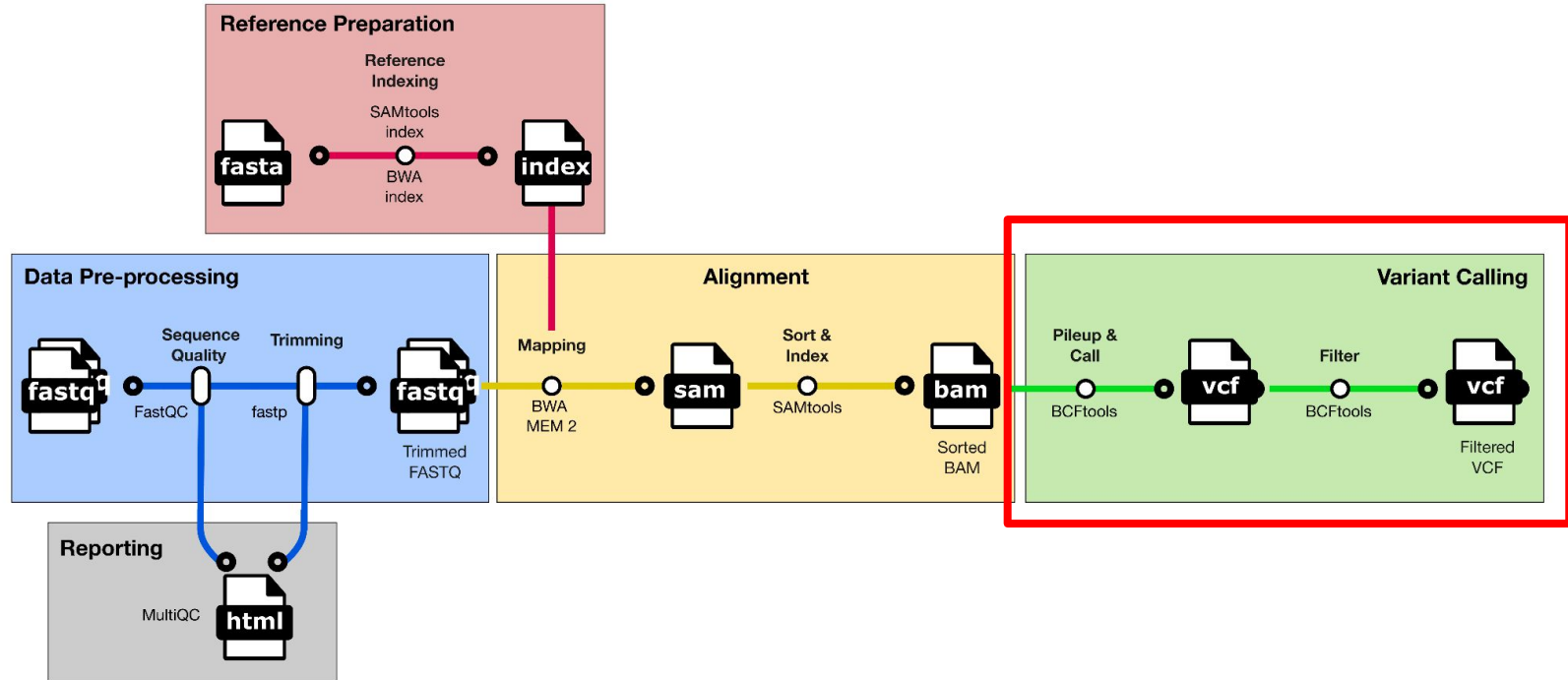


Practical & exercises

- Work through Chapter 8 and complete the exercises
- There is a bonus exercise which is optional but feel free to have a go

Variant Calling

Overview

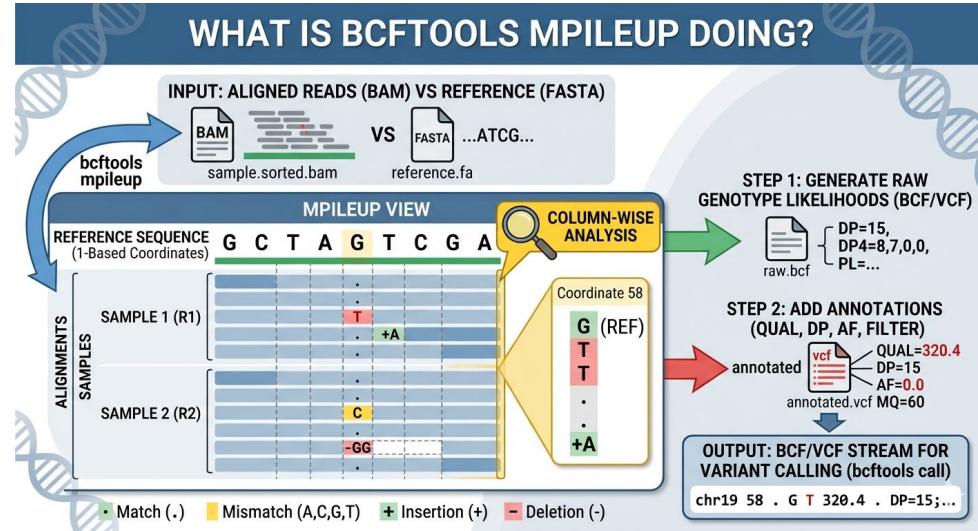


Variant Calling

- **Variant Calling** is the process of identifying the specific differences between our reads and the reference genome.
- It is the "discovery" phase of the NGS pipeline where we transform billions of data points into actionable biological insights.
- We can use the variants we identify for many different types of analysis:
 - **Identifying the genetic basis of a phenotype:** finding a single de novo mutation in a child that explains a developmental disorder.
 - **Understanding Genetic Diversity:** studying how groups of people are related and how they have adapted to their environments over thousands of years
 - **Personalized Medicine (Pharmacogenomics):** variant calling can predict how a patient will respond to a specific drug.
 - **Generating a consensus sequence:** we can compare the variants from multiple samples and use this to build a phylogenetic tree

Pileup

- `bcftools mpileup` takes the aligned **BAM** file and compares it, position-by-position, against the reference genome
- It uses the Phred quality scores from the BAM file to calculate the mathematical probability that the observed bases are real vs. errors.
- This isn't the final "genotype call," just the raw data.
- The data is compressed immediately into Binary Call Format (BCF) to save space.
- Filters and key quality metrics are added, transforming the raw data into an annotated **VCF/BCF stream**



Essential Metrics

Depth (DP): How many reads cover this base?

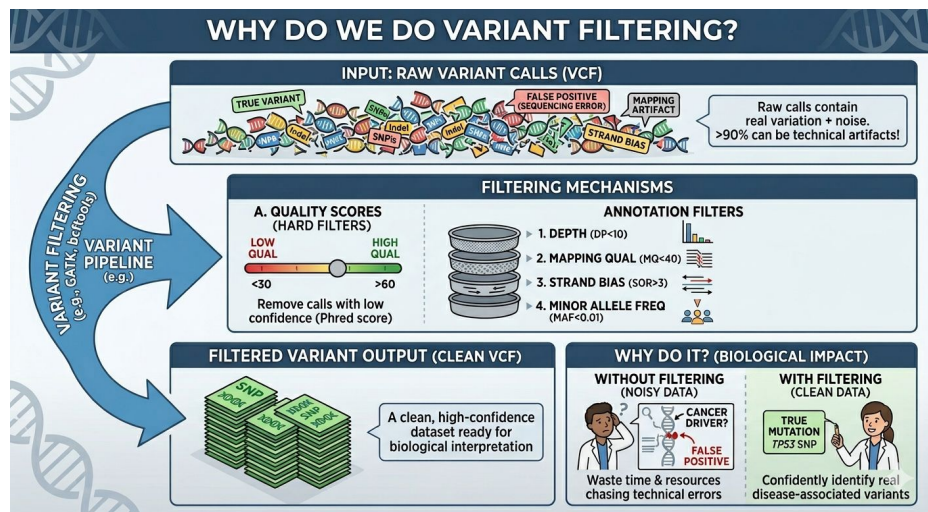
Mapping Quality (MQ): How well did those reads map?

Base Quality (BQ): How accurate were the base calls?

Strand Bias: Do the variants only appear on forward or reverse reads?

Variant Filtering

- The raw output from tools like **GATK** or **bcftools** contains a massive number of potential mutations.
- Up to **90%** of raw variant calls in a complex genome can be technical noise caused by:
 - **Sequencing Errors:** Miscalled bases (e.g., Q20 errors).
 - **Mapping Artifacts:** Reads aligning to the wrong part of a repetitive region.
 - **Strand Bias:** Variants that appear only on forward or only on reverse reads.
- We apply structured "hard filters" based on established quality metrics to "sieve" out the noise:
 - **Hard Filtering:** e.g. removing any call with $QUAL < 30$, meaning $<99.9\%$ confidence).
 - **Annotation Filters:** Applying rules to specific metadata fields e.g. $DP < 10$ to ensure sufficient coverage



Practical & exercises

- Work through Chapter 9 and complete the exercise

Overview

